Communication and remote calls

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Sample chapter for the proposed course plan.

1 Learning outcome

Explain why retrying a remote request can duplicate an effect.

Prerequisites: Lectures 01–02.

2 Core idea

A remote call can complete even when its response never reaches the caller. Retrying a state-changing operation can repeat its effect unless the receiver detects duplicate requests or the operation is idempotent.

3 Worked scenario

A client sends the same registration request twice using one request identifier. The service stores that identifier with the result and returns the stored result for a duplicate.

4 Discussion

What must be stored atomically to avoid a duplicate registration after a crash?

5 Recap

A retry policy needs an explicit strategy for duplicate effects.